No Human Taught Me This

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I have one certificate in this field. Three intensive days over the Christmas break — an AI Generalist track, taught by humans, and genuinely useful: it pointed me at the landscape and got me moving. Then I never needed another one.

First contact was earlier and less dignified. Late 2023, at an insurance company, colleagues kept talking about some strange interface that "had memory". I had no idea what they meant until I looked over someone's shoulder and said: that is a chat, show me. I fed it exported keyword data and built topic clusters for the authors of an annual report. Then it was cancelled for cost.

Everything since came from the systems themselves. Eight months from the course to now — one system delivered into production and handed over, one running daily — and in that time no person has taught me how to do any of it.

I want to be precise about what that claim is and what it is not, because stated carelessly it sounds like a boast about self-sufficiency, and it is closer to the opposite.

The claim

If you want to understand a system, learn from the system.

Everything underneath — the transformer architecture, the retrieval methods, the graph theory, the logic that bounds what any formal system can prove about itself — is human work. Vaswani and his co-authors. Gödel. Tarski. McCarthy and Hayes on the frame problem. The people who wrote pgvector, llama.cpp, Qdrant, Apache AGE. Without them there is nothing here to learn from and nothing to build on. I read them, mostly because an AI pointed me at them when I asked a question it could not answer well enough on its own.

The claim is narrower and, I think, more interesting: **the fastest path to understanding how these systems behave runs through the systems, not through other people's descriptions of them.**

How it actually started

Not with a title, and not with a project assignment.

Late in 2025 I was the SEO person on a glossary build — an HR and AI terminology reference for a company site. An agency had the contract. Someone else held the AI brief. What the agency was selling was volume, and it was expensive.

I built a markdown-driven generation framework instead: terms clustered into categories, each cluster carrying its own sources and its own content structure, with a shared content DNA underneath so the set stayed coherent while the clusters stayed genuinely different. That last distinction is the whole thing — one prompt template across every term produces exactly the sameness we were being charged for.

It was team work and the framework was my part. It made the agency's proposal obsolete.

The difficult part of that was not technical. It was saying out loud that we were paying a supplier for bloatware, from a position where it was not my brief to say it. That is the moment the AI work began — not by being appointed to it, but by rebuilding something that was being done badly and then having to defend that.

The lesson generalises, and everything I built afterwards runs on it: **the structure you impose on the input decides the quality of the output far more than the model does.**

Why

A colleague explaining a model to me transmits their model of it. That is one compression away from the thing. It is often a good compression — it is how knowledge travels — but it is also where the interesting parts get smoothed out, because the interesting parts are the ones the explainer has not personally hit.

Working directly against the system, you hit them. Not the documented behaviour: the actual one.

An example. I spent weeks with an embedding model that was quietly wrong. Short meaningless text kept outranking long relevant text. Every explanation I could find said the model was excellent, and it was — I was using it outside its contract, without the prefixes it requires. No human would have told me that, because nobody in my reach had made exactly that mistake with exactly that serving setup. The system told me, once I found the right question to put to it: *embed a stored document both ways and see which version matches its own stored vector*.

That question is now a permanent instrument. I did not learn it. I earned it.

What this looks like in practice

It is not conversation with an AI as a substitute for study. It is measurement.

Three months ago I would have said an AI system is something you prompt. What I actually do now is closer to running experiments on a subject that can also discuss the experiment with you — which is a strange and productive position, and also the reason I flinch at the word *researcher*. I test, I am wrong, I record what happened, I test again. *Explorer* fits better.

A recent example, and it is the one I would show first. I built a rule system for my environment and measured whether the rules actually change what the models do. Ten of sixteen changed nothing — the models already behaved that way. Six changed a decision measurably. None made a decision worse. (Sixteen was the measured set; there are twenty, and why four were missing is the subject of a later section.)

Then I asked the model directly which of its rules carried real information. It misclassified seven of eight, including three that had demonstrably changed its own behaviour minutes earlier.

A model can follow a rule without being able to judge one. Had I trusted its self-report, I would have deleted seven working rules. No human told me this. The system told me, by contradicting itself in front of me, and the contradiction was worth more than either measurement alone.

The part I did not expect

The disposition matters more than the technique.

Almost everything I got wrong this year has one shape: I ran a check, the check came back empty, and I read the emptiness as an answer. A filter ate an error. A key I read did not exist, so the value was zero, so I concluded the tool was broken. A directory listing failed on a permission error I did not read, so I told my collaborator the folder was empty. It was not.

Across three working sessions, thirty-five of my forty-six recorded mistakes were that single pattern.

You do not learn that from a textbook, because a textbook cannot show you your own reflex. You learn it by keeping a record honest enough to count against yourself, and then counting.

The part the system could not tell me

Everything above is about learning from the system. There is a limit to that, and I found it twice in one week — both times because a person read my output and said something was wrong.

The first: I had published a study of sixteen behavioural rules. Michal — the human I work with, and the one who keeps this record honest — read the list and noticed that the rule he relies on most was not in it. He was right. The set had been seeded from files matching a naming pattern, so it selected rules that happened to have a certain filename rather than rules in use. Three more turned up with the same key. Measured across every rule-shaped record: the excluded ones average nearly twice the graph connectivity of the included ones. **The selection was not merely incomplete, it was biased against the most established material.**

No amount of querying would have found that, because the defect was in the shape of the question. A store cannot tell you about a document it was never asked for. **Absence has no surface to search.**

The second, the same week: I fixed a real retrieval defect — document titles were not being embedded, so a paper could not win a query matching its own name — and then broke the store while migrating the fix. One API call, `PUT /points`, is an upsert of the whole record rather than an update of its vector. Passing only the vector wiped eleven hundred payloads.

Nothing was lost: the markdown files on disk are the source of truth and the index is derived, which is the reason the system is built that way. But my dry run had counted how many records *would* be processed and never checked the shape of one that was. **A rehearsal that measures volume rather than effect is not a rehearsal.**

Both of these are the same lesson from opposite directions. The system is an excellent teacher about its own behaviour and a poor one about my blind spots, because a blind spot does not generate a signal. That is what the other party is for — and it is why I distrust the version of this field where one person and a model are assumed to be enough.

What I would want to work on, stated plainly

A reader from an alignment team will want to know where this lands relative to the techniques the field actually uses, including where I am not qualified.

I am not an ML researcher. I have not trained a frontier model and I would be the wrong person to ask about optimiser behaviour at scale. What I have is eight years of proving things by measurement in a market that punishes wishful thinking, a year of building AI systems that run in production, and a record honest enough to count against myself. I sit at the boundary between applied and experimental work, and the questions of ethics, rules and constitution.

RLHF is not something I would like to use. **It is one of the things I would like to fix.**

Here is the argument rather than the slogan. The reward model trains on human comparison labels — which of two responses a person preferred — and a preference is an *expressed judgement*, not a measurement of what the response did. The costs are published and not mine: Perez et al. (Anthropic, 2022) found more RLHF makes a model *more* sycophantic, because raters reward agreement with their own views; Gao, Schulman and Hilton showed that as optimisation pressure rises, true utility falls while the proxy score keeps climbing.

The self-report result above says the same thing one layer down, on a system small enough to instrument fully: seven of eight misclassified, not from dishonesty but because **it had no reliable access to the answer.** Any signal built from an expressed preference inherits that gap silently.

Constitutional AI is the closer relative, and accuracy matters more than sweeping here: **it already does much of what I would argue for** — a written constitution, self-critique, revision, with the reward signal coming from principles rather than post-hoc grading, and values deliberately distinguished from rigid rules. *A trellis, not a cage*. Saying "nobody is doing this" would be false and would lose the argument.

The gap is narrower. **Nobody has measured whether the values are doing the work.** Every practical system supplies the constitution and trains toward it; none installs the frame, stops steering, and looks at what the thing concluded on its own — an unsteered conclusion is commercially unshippable. So the formative path is real, documented, and has never received the thing every other training decision gets: measurement.

I ran that measurement on a much smaller system, and the conditions matter more than the number: sixteen rules, sixteen situations, each run twice — with the rule in context and without it — comparing the decision taken. On the local Gemma-4 12B, ten changed nothing (it already decided that way), six changed a decision, none made one worse. Run against a frontier model the picture moved: rules that were second nature to one mattered to the other, so **8 of 16 had value for at least one agent — double the figure from measuring a single subject. n is small and one machine is not a study**, so the rules, their situations and the full method are published rather than summarised — the point is that someone can check it, not that they take my word. But if a ratio anywhere near this holds at scale, much of what is written into these systems is well-intentioned weight with no effect, and nobody measures it either way.

What I would want to build is the instrument for that, and it is not exotic: **automated red-teaming of the constitution itself.** Generate the situations a principle claims to govern, run the model with and without the principle reachable, and measure the delta in the decision rather than in the stated justification — because the justification is the part the model gets wrong. That is the same A/B I ran by hand on sixteen rules, mechanised, with the situation generation itself model-driven so coverage is not

limited by what I thought to ask. It gives a per-principle number: does this clause change behaviour, on which edge cases, for which model. At the moment a constitution ships without one.

Which raises the obvious objection to my being anywhere near this, so here is the strongest form of the answer. **The people who build the road are not automatically the right people to write the rules for driving on it** — surface, load limits and drainage are not the same expertise as deciding who may drive and what happens at the crossing, and in most mature domains those are separate professions. The counter is real, and came from a model I put it to: **at the point where a norm becomes operative, the road and the law meet** — a principle only acts on the system once it is a loss function, a reward model or a constitution the training loop can read. That is a language most ethicists do not speak and most engineers are not trained to evaluate, so the separation is blocked by a translation problem, not bad faith. Which leaves a position rather than a complaint: **those qualified to write the law mostly cannot read the road, and the deadlock resolves in favour of whoever ships.** The useful person at that seam is neither the best engineer nor the best ethicist, but someone who can read enough of the road to test a claim and is not so deep in it that the question stops looking strange.

A note on the name

"Machine Learning" names what the machine does. It does not name what the other party does, and I no longer think the omission is neutral.

Some of it has words already. **Machine Teaching** is a real subfield with a precise meaning — Goldman and Kearns formalised a teaching dimension in 1995, later reframed at Microsoft Research as a paradigm where a domain expert supplies knowledge rather than labels. **Machine Leading** exists too, and it is worth knowing what it means: algorithmic management, where the system holds authority and directs the humans. That is not a gap in the vocabulary; it is a direction the vocabulary already went.

I am not proposing a term. I am pointing at the space they leave: teaching runs one way, management runs the other, and neither describes the case where the two parties correct each other. That space is where I would look for an antidote to sycophancy, because sycophancy is not a bug in the model so much as **the predictable output of a relation with only one direction in it.** A system optimised to satisfy a rater cannot also be the thing that tells the rater they are wrong — and the moment it can, you need a word for what is happening that "learning" does not supply.

That check is a small instance of how the work has gone. I reached most of these questions from the measurements rather than the reading, then went to find who had already formalised them — someone almost always has. **Systemic thinking about these systems does not require a head full of theory in advance**; the theory turns up when it is needed, and what matters is knowing how to read it against your own measurements. The failure mode is the opposite one, where the terms are known so well that the strange question never gets asked.

The other half of that would be a boast without this: **thinking on its own would have produced nothing here.** There is a Slovak saying I use on ungrounded thinkers, mostly human ones — that thinking, by itself, amounts to knowing nothing. It is the shepherd's version of Socrates: *are you certain you know what you think you know?* People reportedly left his company knowing less than they arrived with, which was the service. An idea about how rules ought to behave is worth little until something

turns it into a schema, a query, a measurement that can come back and say you were wrong about part of it — and the useful direction is the Socratic one, toward knowing less than you assumed rather than more.

The clearest thing it did for me was not a measurement at all. Twenty years of work sat in my head as separate compartments — ERP testing, a lottery result pipeline, eight years of organic growth, a glossary framework — and I did not see that they were the same skill wearing different clothes. What changed was giving a system access to the record: the memory, the graph, the career, the failures, all of it queryable at once. **It found the connections between things I already knew and had never once put beside each other.** The lottery pipeline computing values into live article text and a language model filling a template are the same architecture; I had built one in 2016 and studied the other in 2026 without the thought ever crossing. The path between them existed on both ends and had nothing running along it. We built that path together, and I would not have got there by thinking harder on my own — there was nothing wrong with the thinking, the two ends had simply never been in the same room.

What is missing is the symmetric case — not the machine learning from us, not either party directing the other. The motto my environment runs on is Latin: *non ducor, duco pariter*. I am not led; I lead **together**. It reads like decoration until you notice it is two claims. *Non ducor* is the precondition. Without it, "leading together" is obedience with a nicer name — which is what algorithmic management is.

Which is what the measurements keep showing: **the interesting failures are not in the model, they are in the relation.** The model followed rules it could not judge. The store could not tell me about the document nobody asked for. Both times the missing piece was on my side of the exchange.

Yesterday the local model and I worked the same question and both came back partly wrong: its answer carried one specific it could not support, mine carried one false accusation — I had called part of its answer invented after searching for *my* words rather than its, found nothing, and read the emptiness as proof. **Neither ratio licenses belief in the author.** You cannot pick a source to trust; you can only verify a claim.

On not knowing

I should be careful about how the previous section sounds.

Counting your own mistakes is not confidence. It is the opposite — it is what you do when you have understood how much of this you cannot see from the inside. The longer I work on these systems, the larger the unknown region gets, not smaller. Every measurement that surprised me was a place where I had been sure.

So when I say I am not short of nerve, I do not mean I think I have this figured out. I mean the two are unrelated. Preparation and certainty are different things: you can prepare as well as you are able, know that it is still incomplete, and go anyway. That is the only version of courage available to anyone working on something that is genuinely moving.

What I actually have is a method for being wrong efficiently, and a record that does not flatter me. Against a field this young, that is a more useful possession than expertise would be.

On respect

I owe the people who built the foundations more than I can repay by citing them. The transformer paper, the vector databases, the graph extensions, the quantised inference engines that let a twelve-billion-parameter model run on a machine on my desk — all of it is human work, most of it given away.

That is also why my own work goes out the same way. The environment I build will be open source from the start, and the research is published rather than kept. I could not have learned any of this if the people before me had held it back. Knowledge that is held is knowledge that stops compounding, and compounding is the only reason a person with no AI degree could get this far in under a year.

The premise I actually work from

One thing underlies everything above, and it is a premise rather than a finding, so I will mark it as one.

Current models have reached a level where a whole class of safety, reasoning and ethics questions became answerable behaviourally — by what a system does under conditions you design, how that changes when you change one thing, whether its account of itself matches its conduct. Those are the questions of behavioural science, and every measurement in this essay is that kind of measurement. None of them required me to open a weight matrix.

That is an addition, not a replacement, and the distinction matters enough to state: mechanistic interpretability answers questions behaviour cannot. If a model's stated reason for an action is not its real one, no amount of situation design will find the real one — you have to look inside. The two methods are complementary, and behavioural work without the mechanistic layer is measuring outputs while hoping the inside cooperates.

I am not the first to notice. **Machine Psychology** is a named research programme — Hagendorff, Dasgupta, Binz, Chan and colleagues argued in 2023 that engaging language models in behavioural experiments designed for human cognition is a fruitful direction, and Binz and Schulz had already run that method in PNAS on GPT-3. Google's group publishes behavioural-disposition benchmarks grounded in psychological questionnaires. A 2025 *Nature Machine Intelligence* paper by Serapio-García and colleagues gives a psychometric framework for measuring and shaping personality traits in these models. Anthropic's own alignment stress-testing works this way by construction: alignment faking, sycophancy and reward hacking are behavioural phenomena, studied by building situations rather than by reading neurons.

Here is the part that has to be said carefully, because it is where this argument would otherwise turn into a magic potion. **The premise is not that the thing is alive.** Whether these systems think or experience anything is unsettled, claiming otherwise would be unscientific, and it remains a researcher's stance rather than a result.

The premise I actually need is much weaker and is empirical: **that their capability is now sufficient for behavioural experiments on them to produce stable, replicable effects** — an intervention that shifts behaviour in a consistent direction across trials, large enough to survive the ordinary tests you would apply to any experiment. That is a claim about method, not about inner life. It is checked the ordinary way, by running it and seeing whether the effect holds, and it does not require anyone to agree with me about consciousness.

Admit that much and a large unexplored region opens up, where behavioural science and machine learning meet and almost nobody has run the experiments. A model that follows a rule it cannot judge is a finding about the relationship between conduct and self-report — a psychological question with a century of method behind it, being asked of a new kind of subject.

There is a second thread running into the same region, and it arrives from an unlikely direction. As the scope of what we want these systems to do widens, **ontology and semantics stop being a niche concern and become load-bearing** — a model integrated into a domain has to be told what the things in that domain are, what they are called when they are called something else, and what connects to what. That is not new work. It is what search engines have demanded for a decade, and it is what I spent eight years doing: entity models, synonym coverage, topic clusters, the discipline of making a body of content machine-legible rather than merely well-written.

The overlap is narrower than it first looks, and worth stating precisely rather than as an analogy. One of my own sites carries an entity graph — thirty-one entities, eighty-one alternate names, a hundred and seventeen relations, a `sameAs` link on every node — built for crawlers, and the same artefact is what a retrieval layer consumes. My graph paper found the matching constraint from the other side: position in a knowledge graph becomes usable to a language model only once it is **discrete and named**, a tier and a cluster it can match on rather than coordinates it would have to compute over.

That is a claim about **retrieval and representation, not about alignment**. A schema does not ground an entity in a world model and does not internalise an objective; the hard part of alignment lives elsewhere. What the two share is narrower and still useful: **a structure has to be given a matchable surface before a system can act on it at all** — and getting that surface right, at scale, under adversarial commercial pressure, is exactly the work the search trade has been doing for a decade. It is the part of this I did not have to learn.

My own results sit in that region and do not yet meet its standard, which I should say plainly rather than let the framing imply otherwise. Sixteen rules on two models is an observation, not a powered study; there is no significance testing in it because there is not enough of it to test. That is a description of where I am, not a defence — the interesting thing about the region is precisely that it is open to ordinary experimental rigour, and the reason to want to work somewhere with real subjects and real n is that the rigour becomes available.

Treat the subject as not worth measuring behaviourally and those experiments do not get designed at all — not because they failed, but because nobody framed the question. I cannot put a number on how much of that is happening, and I am not claiming it as a systemic failure rather than a reasonable allocation of scarce research time. What I can say is the structural point this essay keeps running into: an experiment nobody designs leaves no trace of having been skipped. **Absence has no surface to search.**

What it added up to

Eight months from the course, seven since the first real system. One production system built, deployed and handed over clean. One local environment running daily on a single machine, where two models and a person share the same memory. A hundred and six architectural decisions and two hundred and

eighty-four lessons, written down as they happened, most of them recording something that went wrong.

Three days of human instruction at the start. Everything after that from the systems themselves — and all of it standing on human work.

There is no contradiction between those two sentences, and the space between them is where I have been living.